

Objectives

1. Describe and compare the pathophysiology of **pulmonary embolism**, pulmonary **hemorrhage** and **infarct**
2. Discuss the etiology, pathophysiology, lesions, and grading of **pulmonary hypertension**.
3. Interpret the pathophysiology and associated pulmonary lesions of **Goodpasture syndrome**.
4. Specify how the pathophysiology and associated pulmonary lesions of **granulomatosis with polyangiitis** (aka Wegener granulomatosis) differ from similar diseases.

Pulmonary embolisms (PE)

- 3rd most common acute cardiovascular condition worldwide
 - Rare in pediatrics, common after age **75 years**
 - Also, risk for **inpatients** > general population
- **Most** clots occluding large arteries → **embolic**, usually **deep leg vein thrombi** (95%), usually **femoral vein**
 - Rarely have infected (septic infarct), **fat**, air, amniotic fluid or **neoplastic** embolus
 - Example: Small bone **marrow** emboli after chest compressions during **resuscitation** (often in those who die)
- **Usually** in patients with **predisposing conditions** increasing the risk of thrombosis:
 - **Primary** hypercoagulable states: Factor V Leiden, antiphospholipid syndrome, prothrombin **mutations**
 - **Secondary** hypercoagulable states: Cardiac disease, obesity, **cancer**, **pregnancy**, oral **contraceptives**, **post-op**, **immobilization**/bed rest, patients with hip **fractures** (particularly high risk), indwelling central venous lines (nidus for right atrial thrombi formation, can embolize to lungs)
- **Virchow's Triad** (general factors contributing to thrombosis): **Hemodynamic changes** (stasis, turbulence), **endothelial injury**, **hypercoagulability**

PE: Clinical findings

- 60-80% of PE clinically **silent**
 - **Small** emboli may be **silent** or only have **transient** chest pain and cough
 - ~5% cause **acute cor pulmonale**, shock, or **death** (typically large “saddle emboli”)
 - During CPR, patients frequently have electromechanical dissociation (**EMD**- EKG has rhythm, but no pulses palpated because no blood is entering the pulmonary artery circulation)
 - If patient **survives**, condition may clinically **mimic MI** (severe chest pain, dyspnea, and shock)
 - In remaining patients with symptomatic PE, cause symptoms of **ventilation-perfusion (V/Q) mismatch** (when airflow and blood flow in the lungs are not balanced), and/or pulmonary **infarction** - particularly **dyspnea** and **pleuritic** chest pain
 - Most commonly find: Dyspnea, pleuritic pain, **pleural friction rub** (**overlying fibrinous pleuritis**) and **cough**; may have **fever**, **hemoptysis** (with emboli producing infarction), and calf or thigh swelling or pain (half of cases)*
- Recurrent small emboli can give rise to **pulmonary hypertension**

*A positive Homans's sign (calf pain with dorsiflexion of the foot) may be associated with the presence of thrombosis

PE: Prevention/treatments

- After initial acute insult, esp. **young** patients, emboli often **resolve** via contraction & fibrinolysis
- If **unresolved**, multiple small emboli may lead to **pulmonary hypertension & chronic cor pulmonale**
- High risk of **recurrence** → recurrent embolism may → **pulmonary hypertension & cor pulmonale**
- Important to note is that a **small PE may predict a larger one**
- **Prophylactic** therapy (to prevent blood clots like DVT)
 - **Early ambulation** postoperative & postpartum patients
 - Compression **stockings**/hose
 - **Anticoagulation in high-risk** individuals
- Treatment includes **anticoagulation & supportive** measures
 - If risk of recurrence & anticoagulation contraindicated: **Inferior vena cava filter**

Pulmonary hypertension definition: Mean **pulmonary artery pressure ≥ 25 mm Hg at rest**

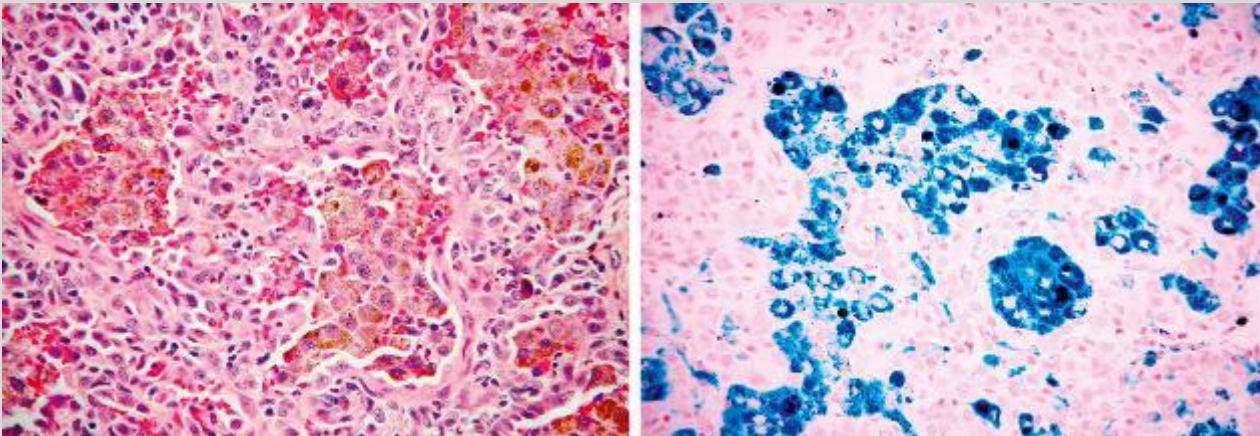
- Direct disease of lung arteries
- PH is a group category based on underlying mechanisms, **WHO classified** PH into five groups that have diverse **causes** even within each group:
 1. Primary pulmonary hypertension (aka **idiopathic** pulmonary arterial hypertension, PAH) – diverse collection, all mostly effect **small** pulmonary muscular **arteries**
 2. Due to **left heart failure** - (most frequent cause of overall PH)
 3. Due to **lung diseases** and/or **hypoxia**
 4. Chronic **thromboembolic** pulmonary hypertension and other **obstructions**
 5. Unclear and/or **multifactorial** mechanisms

Pulmonary hypertension: Clinical findings & treatments

- Signs & symptoms in all types typically evident **only in advanced disease**
 - Early, mild, subtle signs: Shortness of breath during exercise or slight tiredness
- In **idiopathic** disease, usually presents w/ **dyspnea & fatigue**, some have **anginal** chest pain → severe **respiratory distress**, **cyanosis**, syncope, peripheral **edema**, jugular vein distention (**JVD**), right ventricular hypertrophy (**parasternal lift**) with severe disease → death from decompensated **cor pulmonale**, often with superimposed thromboembolism & pneumonia, ensues within 2-5 years in ~80% of patients
- In general PH is most common in **older adults** (often > 65 y.o., with ↑age & underlying heart or lung disease) & women slightly > men
 - Sub-types like **PAH** frequently (60-80%) in **younger** 30-60 y.o. & middle-aged women, rarely children
- Treatment depends on **underlying cause**
 - Vasodilators used in PAH
 - Secondary disease: Therapy is directed at trigger (ex., thromboembolic disease or hypoxemia)
 - Lung transplantation: Definitive treatment (select patients)

Diffuse pulmonary hemorrhage syndromes include

- **Goodpasture syndrome**
- Vasculitis-associated (includes hypersensitivity angiitis, **polyangiitis with granulomatosis**, lupus)
- **Idiopathic pulmonary hemosiderosis**
- Others (ex. CHF)



Diffuse **pulmonary hemorrhage**: Intra-alveolar spaces with large numbers of **macrophages laden** with phagocytosed red cells and hemoglobin breakdown product **hemosiderin (siderophages)** - **Prussian-blue** stain identifies **iron** derived from hemoglobin in macrophages

Robbins, Cotran & Kumar Pathologic Basis of Disease, 11th ed., 2026, Ch. 15, fig 15.26

Goodpasture syndrome involvement of the lung is an anti–glomerular basement membrane antibody disease with pulmonary involvement:

- Uncommon **autoimmune** disease: **Kidney & lung injury** is caused by **circulating autoantibodies** against **noncollagenous** domain of the **$\alpha 3$ chain** of **collagen IV**
- Most cases in **teens or 20s**, **male** preponderance (unlike most autoimmune diseases) & majority active **smokers**
- Called anti–glomerular basement membrane disease if only renal disease

Goodpasture syndrome

- Pathogenic antibodies cause **inflammatory destruction** of **basement membranes** in **lung** alveoli and kidney glomeruli
 - Presents with necrotizing hemorrhagic **interstitial pneumonitis** & usually **rapidly progressive glomerulonephritis**
- **Trigger** that initiates production of anti–basement membrane antibodies is **unknown**
- As with other autoimmune disorders, there is an association with **HLA subtypes** (e.g., HLA-DRB1*1501 & DRB1*1502)

Goodpasture syndrome: Clinical findings & treatments

- **Usually begins** with respiratory symptoms (**hemoptysis**), CXR shows evidence of **consolidations**
- Soon **after respiratory** symptoms: **Glomerulonephritis** manifestations occur and lead to **rapid progressive renal failure**
- Most common cause of death is **uremia**
- Rx: **Corticosteroids**, Plasmapheresis & **Cyclophosphamide**
- Intensive **plasmapheresis** removes anti–basement membrane antibodies and other possible mediators of immunologic injury
- Simultaneous **immunosuppressive therapy** inhibits further antibody production, improving both lung hemorrhage and glomerulonephritis

Granulomatosis with polyangiitis (GPA)

- **Autoimmune** disease that **mostly** involves **upper respiratory tract** and/or **lungs**
 - **c-ANCA 95% (PR3-ANCA)**
- **Hemoptysis** common presenting symptom
- Childhood to middle age, peak incidence ~ 40 y.o., M>F
- A transbronchial lung biopsy might provide the only tissue available for **diagnosis**
 - Important histologic features are **capillaritis** and scattered, poorly formed **granulomas**
 - Characteristically **necrotizing vasculitis** with the following:
 - Acute **necrotizing granulomas** of **upper respiratory tract** and/or **lungs**
 - Granulomas have a **geographic** pattern of **necrosis**
 - Necrotizing or granulomatous **vasculitis** of **small-medium** sized **vessels**, mostly in lungs
- If **renal disease** is present: Proliferative or necrotizing, focal, crescentic **glomerulonephritis**

GPA: Pathogenesis & treatment

- Most likely T cell–mediated **hypersensitivity** response to normally “**innocuous**” **inhaled microbial** or **other environmental agents**
 - Supported by presence of **granulomas** & dramatic **response to immunosuppressive therapy**
- **Necrotizing medium-sized & small vessel vasculitis** involving:
 - **Upper respiratory tract**
 - **Lungs**
 - **Renal vessels**
- Treatment is similar to Goodpasture syndrome

Idiopathic pulmonary hemosiderosis

- **Rare** disorder with **intermittent**, **diffuse** alveolar hemorrhage, mostly in young **children**
 - M>F, peak incidence ~27 years
- **Cause & pathogenesis are unknown** - But, **favorable response** to long-term **immunosuppression** with prednisone and/or azathioprine indicates that an **immunologic** mechanism may be involved
- Defined by **recurrent alveolar bleeding w/o** an identifiable **cause**
- Usually **presents** with an **insidious** onset of productive **cough**, **hemoptysis**, and **anemia** associated with **diffuse bilateral pulmonary infiltrates**
- **Diagnosis of exclusion** - rule out all other causes first

Summary

- Pulmonary **embolism/infarct** – pulmonary infarcts are usually **hemorrhagic** because of dual supply of lungs
- Pulmonary **hypertension** – can be genetic in some cases (**Primary idiopathic**) associated with BMPR2 germline mutation; for other causes see **WHO** classification
- **Goodpasture** syndrome – affects **lungs & kidneys**; **autoimmune**, usually **male**; circulating **auto-antibodies** against **collagen IV**
- **Granulomatosis with polyangiitis** – autoimmune, **95% PR3-ANCA**; **necrotizing vasculitis with triad** of necrosis, granulomas & vasculitis of upper respiratory tract and/or lungs small to medium-sized vessels (mostly in lungs), and **renal** disease (**glomerulonephritis**)
- **Idiopathic pulmonary hemosiderosis** – rare, **children**; **intermittent** diffuse alveolar hemorrhage & hemosiderosis, siderophages, necrosis of alveolar walls, responds to **immunosuppressants**, **Dx of exclusion**